

GitHub Repository

INGESTION & PROCESSING LAYER

Repository Ingestion (GitPython)

- Commit History Extraction
- Metadata Extraction
- README Parsing
- Directory Tree Indexing

Temporal Snapshot Builder

TEMPORAL RAG STACK

Sentence Transformer (all-MiniLM-L6-v2)

ChromaDB Temporal Vector Store

Temporal Retriever

- Semantic Search
- Chronological Filtering
- Query Classification
- Positional Resolution

ANALYTICS & REASONING ENGINE

Ollama Local LLM (Qwen2.5-Coder 7B)

Reasoning Engine

- Repository Chat
- Causal Reasoning (Why)

Repository Intelligence Engine

- Repository Health Score
- Developer Bus Factor
- Evolution Milestone Timeline
- Logical File Coupling
- Developer Analytics
- Risk Hotspots & Churn
- Forensic Bug Origin
- Co-Evolution

React Dashboard